



NORMANHURST BOYS HIGH SCHOOL

MATHEMATICS ADVANCED

2025 Year 12 Course Assessment Task 4 (Trial HSC Examination)

Friday, 1 August 2025

General instructions

- Working time – 3 hours.
(plus 10 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- NESA approved calculators may be used.
- Attempt **all** questions.

SECTION I

- Mark your answers on the answer grid provided (on page 32)

SECTION II

- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

NESA STUDENT #:

Class (please ✓)

☐ 12MAA.1 – Mr Tran

☐ 12MAX.1 – Miss J. Kim

☐ 12MAX.2 – Miss Chaussivert

☐ 12MAX.3 – Miss Lee

☐ 12MAX.4 – Mr Lam

Marker's use only.

QUESTION	1-10	11-14	15-17	18-21	22-24	25-27	28-29	Total
MARKS	$\overline{10}$	$\overline{15}$	$\overline{14}$	$\overline{16}$	$\overline{14}$	$\overline{17}$	$\overline{14}$	$\overline{100}$

Section I

10 marks

Attempt Question 1 to 10

Allow approximately 15 minutes for this section

Mark your answers on the answer grid provided (labelled as page 32).

Questions	Marks
1. What is the period of the function $f(x) = 2 \cot(3x)$?	1
(A) $\frac{\pi}{3}$	(C) $\frac{2\pi}{3}$
(B) $\frac{\pi}{2}$	(D) π
2. For a function $f(x)$, what type of dilation is required to sketch $y = f(4x)$?	1
(A) Vertical dilation by a factor of $\frac{1}{4}$	
(B) Horizontal dilation by a factor of $\frac{1}{4}$	
(C) Vertical dilation by a factor of 4	
(D) Horizontal dilation by a factor of 4	
3. Which of the following are the asymptote(s) of the graph of $y = \log(x - 1) + 2$?	1
(A) $x = -1$ only	(C) $y = 2$ only
(B) $x = 1$ only	(D) $x = 1$ and $y = 2$
4. What is the solution of $3^x = 5$?	1
(A) $\frac{\log 5}{3}$	(C) $\frac{\log 5}{\log 3}$
(B) $\frac{3}{\log 5}$	(D) $\log\left(\frac{5}{3}\right)$

5. Which of the following is the domain of the function 1

$$f(x) = \frac{1}{\sqrt{x^2 - 4}}$$

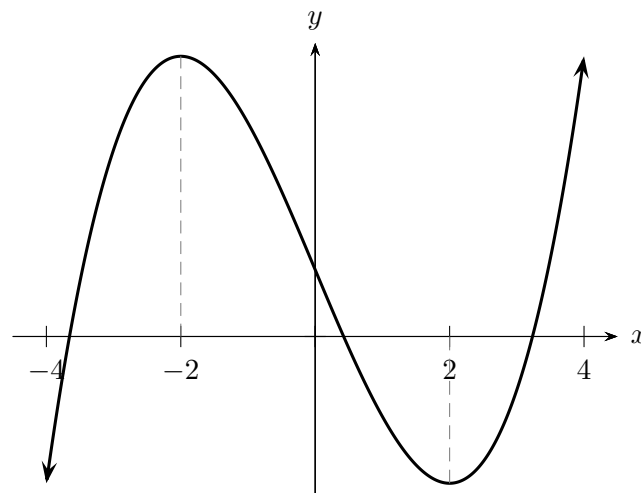
- (A) $[-2, 2]$ (C) $(-2, 2)$
 (B) $(-\infty, -2] \cup [2, \infty)$ (D) $(-\infty, -2) \cup (2, \infty)$

6. Which of the following represents the range of the function 1

$$f(x) = -x^2 - 2x - 3 \text{ for } x \in (-2, 1]$$

- (A) $[-6, -3]$ (C) $[-6, -3]$
 (B) $[-6, -2]$ (D) $[-6, -2]$

7. From the graph of $y = f(x)$, when is $f'(x)$ negative? 1



- (A) $x < -2$ or $x > 2$ (C) $x \leq -2$ or $x \geq 2$
 (B) $-2 < x < 2$ (D) $-2 \leq x \leq 2$
8. Find all the values of k , such that the equation $x^2 + (4k + 3)x + \left(4k^2 - \frac{9}{4}\right) = 0$ 1
 has two real solutions for x , where one of the solutions is positive and the other is negative.

- (A) $k > -\frac{3}{4}$ (C) $-\frac{3}{4} < k < \frac{3}{4}$
 (B) $k > \frac{3}{4}$ (D) $k < -\frac{3}{4}$ or $k > \frac{3}{4}$

Examination continues overleaf...

9. Evaluate $\lim_{x \rightarrow 2} \frac{2-x}{1-\sqrt{x-1}}$ 1
- (A) -2 (B) $\frac{1}{2}$ (C) 2 (D) ∞

10. Using the table below, what is the maximum possible value for $E(X)$? 1

x	-1	0	1	2
$P(X = x)$	k^2	$3k$	k	$-k^2 - 4k + 1$

- (A) $\frac{2}{3}$ (B) 1 (C) 2 (D) $\frac{73}{12}$

Section II

90 marks

Attempt Question 11 to 29

Allow approximately 2 hours and 45 minutes for this section

Write your answers in the space provided.

Question 11 (3 marks)

Given $f(x) = \sqrt{x}$ and $g(x) = 25 - x^2$.

(a) Find $f(g(x))$. 1

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(b) Find the domain and range of $f(g(x))$. 2

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Question 12 (3 marks)

Solve for x : 3

$$2 \ln(x) - \ln(x + 3) = \ln\left(\frac{1}{2}\right)$$

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Examination continues overleaf...

Question 13 (6 marks)

- (a) The function $y = |x|$ is subjected to the following transformations. For each of the transformations, write the correct equation that arises.
- i. Vertical dilation by a factor of 2. **1**
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 - ii. Horizontal translation of the equation in part (i) to the left by 1 unit. **1**
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 - iii. Horizontal dilation of the equation in part (ii) by a factor of $\frac{1}{2}$. **1**
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- (b) Hence or otherwise, find the value of a , where $a > 0$ such that **3**

$$\int_{-\frac{3}{4}}^a (2|2x+1| - 1) dx = 0$$

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Question 14 (3 marks)

Initially there are 250 litres of water in a tank. Water starts to flow into the tank.

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The rate of increase of the volume of water in litres is given by

$$\frac{dV}{dt} = 250 - 7.5t$$

where t is the time in hours.

Find the volume of water in the tank (correct to the nearest litre) when $\frac{dV}{dt} = 0$.

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Examination continues overleaf...

Question 15 (6 marks)

For the function:

$$f(x) = xe^x$$

- (a) Find the coordinates of the stationary point(s) and determine the nature, the coordinates of the point(s) of inflection, and sketch the curve of $f(x)$, showing all key features. **5**

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- (b) Hence or otherwise, find the value(s) of k , where k is a real number for which the equation $xe^x = k$ has 2 solutions.
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Question 16 (4 marks)

A random variable X has this probability distribution:

x	1	2	3	4	5
$P(X = x)$	0.2	0.1	0.25	0.3	0.15

- (a) Find $P(X \geq 2|X < 4)$.
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- (b) Find the variance of X , $\text{Var}(X)$.
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Examination continues overleaf...

Question 17 (4 marks)

Joe plays soccer, a sport where a team can win, lose or draw.

Joe is prone to injury, and misses 20% of his team's games.

When Joe plays, his team has a 77% chance of winning and a 10% chance of losing.

When Joe does not play, his team has a 53% chance of winning and a 25% chance of losing.

- (a) Calculate the probability that Joe's team loses any given game. **2**

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- (b) Given that Joe's team does not lose in a particular match, what is the probability that Joe played in that match? **2**

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Question 18 (4 marks)

Prove the following identity:

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$$\frac{\cos \theta}{\cos \left(\frac{\pi}{2} - \theta\right)} (1 + \tan^2 \theta) + \frac{\sin \theta}{\cos \theta} \left(1 + \tan^2 \left(\frac{\pi}{2} - \theta\right)\right) = 2 \operatorname{cosec} \theta \sec \theta$$

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Examination continues overleaf...

Question 19 (3 marks)

The number of concert tickets sold, N , is given by

$$N = 1000 \times 1.6^{kt}$$

Where t is the number of weeks after being available for sale, and k is a constant. Initially, 1000 tickets were sold, and after 2 weeks, the total number of tickets sold was doubled.

- (a) Calculate the value of k , leaving your answer in exact form.

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- (b) At what rate was the number of tickets sold increasing after 4 weeks? Give your answer to the nearest whole number.

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Question 20 (6 marks)

A fisherman's boat is trying to reach the open sea, however, it is heavily affected by the ocean tides. To reach the open sea, the boat must cross a sandbar which requires the water to be at least 4 metres deep. On a particular day, high tide occurred at midnight, and the depth, D , of water over the sandbar on the ensuing day is given by $D = 2 \cos\left(\frac{4\pi}{25}t\right) + 3$, where t is the number of hours after midnight.

- (a) Sketch the graph of $D = 2 \cos\left(\frac{4\pi}{25}t\right) + 3$, for $0 \leq t \leq 25$, showing all key features. **3**

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- (b) Find the times when the fisherman would not be able to cross the sandbar in his boat. **3**

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Examination continues overleaf...

Question 21 (3 marks)

Show that $\int_{-1}^1 \frac{1}{1+e^{-x}} dx = 1$.

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Question 22 (5 marks)

It is known that $T_1 = 1 + \cos A$, $T_2 = \frac{1}{2}$ and $T_3 = 1 - \cos A$ form the first three consecutive terms of a geometric sequence, with $T_1 > T_2 > T_3$.

- (a) Find the value of A , given $\angle A$ is an angle within a triangle.

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- (b) Using the value of A found in part (a), show that the series

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$$(1 + \cos A) + \frac{1}{2} + (1 - \cos A) + \dots$$

has a limiting sum.

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Examination continues overleaf...

Question 23 (3 marks)

If $y^2 = (2m - y)(2n - y)$, show that $\frac{1}{m}, \frac{1}{y}, \frac{1}{n}$ (in this order) forms the terms of an arithmetic sequence. **3**

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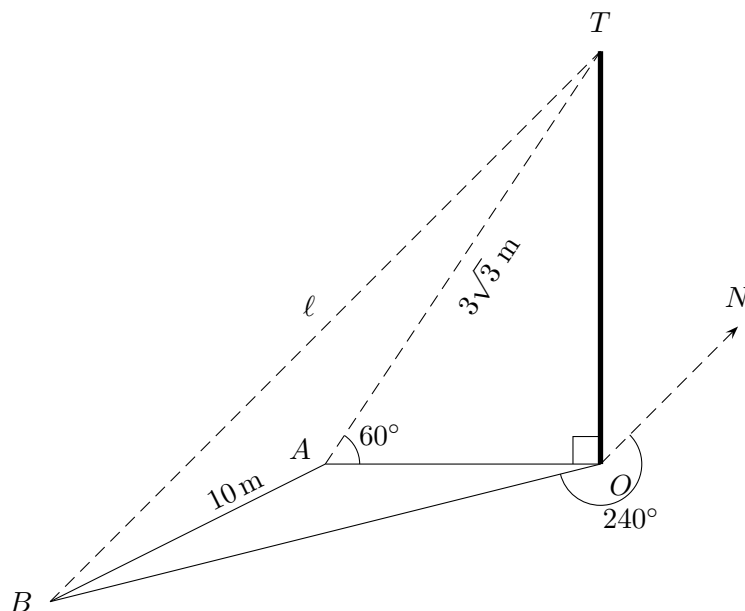
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A tall pole OT is held in place vertically by ropes anchored by two pegs in the ground 10 metres apart at A and B as shown. A is due west of O , and B is at a bearing of 240°T from O . The rope connecting the top of the pole T to the peg at A is $3\sqrt{3}$ metres long and makes an angle of 60° with the ground.



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- (b)

Hence or otherwise, find the length of the rope, ℓ , connecting T and B .
Give your answer to the nearest metre.
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Examination continues overleaf...

Question 25 (3 marks)

The line $y = mx$ is a tangent to the curve $y = \ln(kx)$, where $k > 0$.
Find the value of m , in terms of k .

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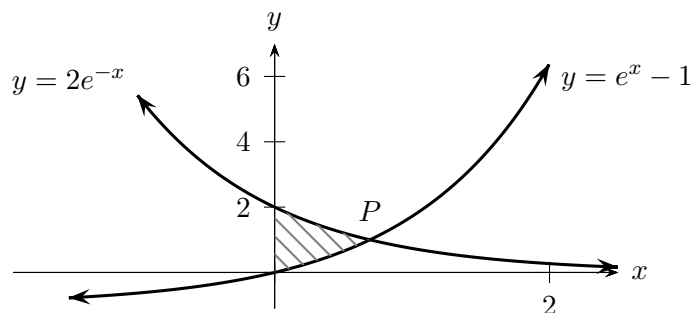
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Question 26 (6 marks)

Two curves $y = 2e^{-x}$ and $y = e^x - 1$ intersect at point P .



- (a) Show that the coordinates of P are $(\ln 2, 1)$.

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- (b) Hence, calculate the shaded area in simplest exact form.

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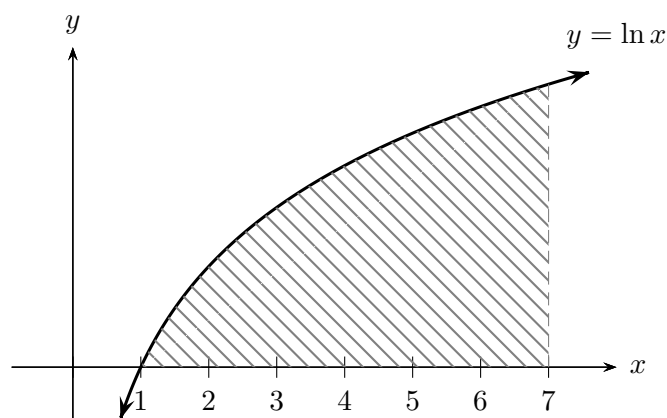
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Question 27 (8 marks)

The diagram below shows the graph of $y = \ln x$. The region bounded by the graph, the x -axis and the line $x = 7$ has been shaded.



- (a) Using the trapezoidal rule with 4 function values, find the approximate shaded area in exact form. **2**

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- (b) Find $\frac{d}{dx}(x \ln x)$. **2**

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- (c) Hence, or otherwise, find $\int_1^7 \ln x \, dx$.

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- (d) Deduce that $\ln 7 \approx \frac{1}{3} (\ln 3 + \ln 5) + 1$.

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Examination continues overleaf...

Question 28 (7 marks)

The acceleration of a particle moving along the x -axis is given by $\ddot{x} = 4 \sin 2t$, where x is the displacement of the particle from the origin in metres, and t is the time in seconds. Initially, the particle is at the origin moving to the left at 1 metre per second.

- (a) Show that the velocity of the particle is given by $\dot{x} = 1 - 2 \cos 2t$. **2**

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- (b) Find the time when the particle first comes to rest. **2**

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- (c) Find the distance travelled by the particle in the first $\frac{\pi}{2}$ seconds.

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Examination continues overleaf...

(b) Find the maximum perimeter of the rectangle. Leave your answer in exact form. 4

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End of paper.

Suggested Band 6 Responses

Section I

1. (A) 2. (B) 3. (B) 4. (C) 5. (D)
6. (B) 7. (B) 8. (C) 9. (C) 10. (C)

Section II

Question 11 (3 marks) (Tran)

(a) (1 mark)

✓ [1] for correct answer

$$f(g(x)) = \sqrt{25 - x^2}$$

(b) (2 marks)

✓ [1] for correct domain

✓ [2] for correct range

$$D_{f(g(x))} = \{x : x \in [-5, 5]\}$$

$$R_{f(g(x))} = \{y : y \in [0, 5]\}$$

Question 12 (3 marks) (Tran)

✓ [1] for equating argument of logarithms

✓ [2] for obtaining all possible solutions

✓ [3] for correct answer

$$\ln(x^2) - \ln(x+3) = \ln\left(\frac{1}{2}\right)$$

$$\ln\left(\frac{x^2}{x+3}\right) = \ln\left(\frac{1}{2}\right)$$

$$\frac{x^2}{x+3} = \frac{1}{2}$$

$$2x^2 = x + 3$$

$$2x^2 - x - 3 = 0$$

$$(2x-3)(x+1) = 0$$

$$x = \frac{3}{2} \text{ or } -1$$

Since $x > 0$:

$$x = \frac{3}{2}$$

Question 13 (6 marks) (Tran)

- (a) i. (1 mark)
 ✓ [1] for correct answer

$$2f(x) = 2|x|$$

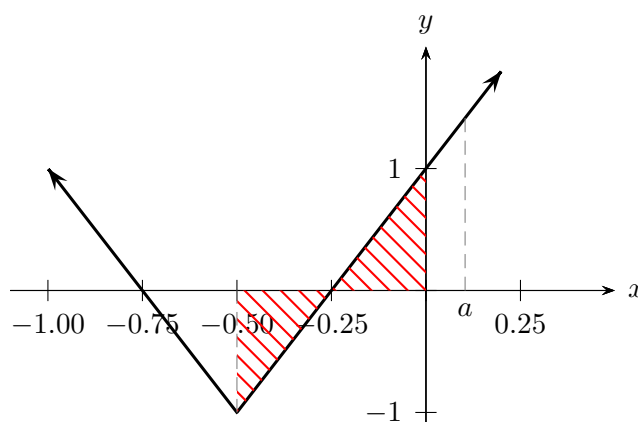
- ii. (1 mark)
 ✓ [1] for correct answer

$$2f(x+1) = 2|x+1|$$

- iii. (1 mark)
 ✓ [1] for correct answer

$$2f(2x+1) = 2|2x+1|$$

- (b) (3 marks)
 ✓ [1] for obtaining the correct integral or equivalent merit
 ✓ [2] for obtaining $16a^2 + 8a - 1 = 0$ or equivalent merit
 ✓ [3] for correct answer



$$\begin{aligned}
\int_{-0.75}^{-0.5} 2|2x+1| - 1 \, dx &= \frac{1}{2} \times \frac{1}{4} \times 1 \\
&= \frac{1}{8} \\
\int_0^a 2|2x+1| - 1 \, dx &= \frac{1}{8} \\
\int_0^a 4x+1 \, dx &= \frac{1}{8} \\
\left[2x^2 + x \right]_0^a &= \frac{1}{8} \\
2a^2 + a &= \frac{1}{8} \\
16a^2 + 8a - 1 &= 0 \\
a &= \frac{-8 \pm \sqrt{8^2 - 4 \times 16}}{2 \times 16} \\
&= \frac{-8 \pm 8\sqrt{2}}{32} \\
&= -\frac{1}{4} + \frac{\sqrt{2}}{4} \quad (a > 0)
\end{aligned}$$

Alternative Method:

$$\begin{aligned}
\int_{-0.75}^{-0.25} 2|2x+1| - 1 \, dx &= \frac{1}{2} \times \frac{1}{2} \times 1 \\
&= \frac{1}{4} \\
\int_{-0.25}^a 2|2x+1| - 1 \, dx &= \frac{1}{4} \\
\frac{1}{2} \times \left(a + \frac{1}{4} \right) \times (2(2a+1) - 1) &= \frac{1}{4} \\
\left(a + \frac{1}{4} \right) \times (4a+1) &= \frac{1}{2} \\
4a^2 + 2a + \frac{1}{4} &= \frac{1}{2} \\
16a^2 + 8a - 1 &= 0 \\
a &= \frac{-8 \pm \sqrt{8^2 - 4 \times 16}}{2 \times 16} \\
&= \frac{-8 \pm 8\sqrt{2}}{32} \\
&= -\frac{1}{4} + \frac{\sqrt{2}}{4} \quad (a > 0)
\end{aligned}$$

Question 14 (3 marks) (Tran)

- ✓ [1] for obtaining $V(t)$ equation or equivalent merit
- ✓ [2] for finding the value of t or equivalent merit
- ✓ [3] for correct answer

$$\begin{aligned}
 V(t) &= 250t - 3.75t^2 + C \\
 \text{Since } V(0) &= 250 \\
 C &= 250 \\
 V(t) &= 250t - 3.75t^2 + 250 \\
 \text{When } \frac{dV}{dt} &= 0 \\
 250 - 7.5t &= 0 \\
 t &= \frac{100}{3} \\
 V\left(\frac{100}{3}\right) &= 250\left(\frac{100}{3}\right) - 3.75\left(\frac{100}{3}\right)^2 + 250 \\
 &= 4416.66 \dots \\
 &= 4417 \text{ Litres}
 \end{aligned}$$

Question 15 (6 marks) (J. Kim)

(a) (5 marks)

- ✓ [1] for finding coordinates of stationary point
- ✓ [2] for showing stationary point is local minimum or equivalent merit
- ✓ [3] for proving there exist a point of inflection or equivalent merit
- ✓ [4] for correct shape/scale of graph of $f(x)$ or equivalent merit
- ✓ [5] for correct solution, all key features were labeled the graph of $f(x)$

$$\begin{aligned}
 f'(x) &= e^x + xe^x \\
 &= e^x(x+1)
 \end{aligned}$$

Let $f'(x) = 0$

$$\begin{aligned}
 x &= -1 \quad (e^x \neq 0) \\
 f(-1) &= -\frac{1}{e}
 \end{aligned}$$

x	-2	-1	0
$f'(x)$	-0.14	0	1
$f(x)$	$\searrow$	$-\frac{1}{e}$	$\nearrow$

$$\therefore \left(-1, -\frac{1}{e}\right) \text{ is a local minimum}$$

$$\begin{aligned}
 f''(x) &= e^x + e^x + xe^x \\
 &= e^x(x+2)
 \end{aligned}$$

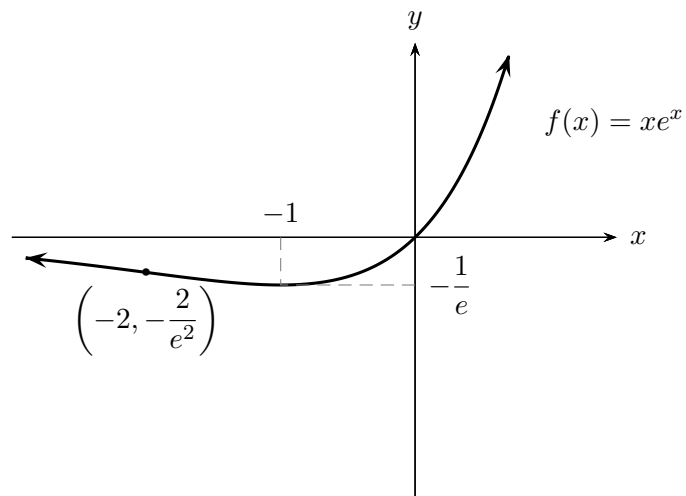
Let $f''(x) = 0$

$$x = -2 \quad (e^x \neq 0)$$

$$f(-2) = -\frac{2}{e^2}$$

x	-3	-2	0
$f'(x)$	-0.05	0	2
$f(x)$	$\frown$	$-\frac{2}{e^2}$	$\smile$

$\therefore \left(-2, -\frac{2}{e^2}\right)$ is a point of inflection



(b) (1 mark)

✓ [1] for correct answer

$$-\frac{1}{e} < k < 0$$

Question 16 (4 marks) (J. Kim)

(a) (2 marks)

- ✓ [1] for identifying all possible outcomes or equivalent merit
- ✓ [2] for correct answer

$$\begin{aligned}
 P(X \geq 2 | X < 4) &= \frac{P(X \geq 2 \cap X < 4)}{P(X < 4)} \\
 &= \frac{0.1 + 0.25}{0.2 + 0.1 + 0.25} \\
 &= \frac{7}{11}
 \end{aligned}$$

(b) (2 marks)

- ✓ [1] for obtaining $E(X^2)$ or equivalent merit
- ✓ [2] for correct answer

$$\begin{aligned}
 E(X^2) &= 0.2 + 2^2 \times 0.1 + 3^2 \times 0.25 + 4^2 \times 0.3 + 5^2 \times 0.15 \\
 &= 11.4
 \end{aligned}$$

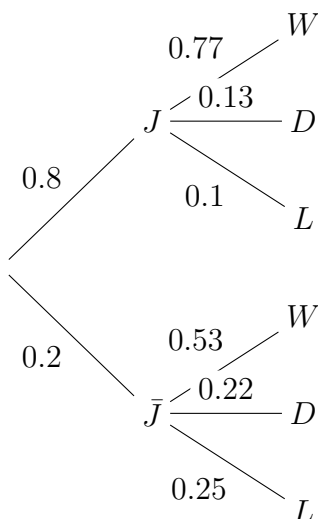
$$\begin{aligned}
 E(X) &= 0.2 + 2 \times 0.1 + 3 \times 0.25 + 4 \times 0.3 + 5 \times 0.15 \\
 &= 3.1
 \end{aligned}$$

$$\begin{aligned}
 \text{Var}(X) &= 11.4 - 3.1^2 \\
 &= 1.79
 \end{aligned}$$

Question 17 (4 marks) (J. Kim)

(a) (2 marks)

- ✓ [1] for identifying all possible outcomes or equivalent merit
- ✓ [2] for correct answer



$$\begin{aligned}
 P(L) &= P(JL) + P(\bar{J}L) \\
 &= 0.8 \times 0.1 + 0.2 \times 0.25 \\
 &= 0.13
 \end{aligned}$$

(b) (2 marks)

- ✓ [1] for identifying all possible outcomes or equivalent merit
- ✓ [2] for correct answer

$$\begin{aligned}
 P(J|\bar{L}) &= \frac{P(J \cap \bar{L})}{P(\bar{L})} \\
 &= \frac{0.8 \times (0.77 + 0.13)}{1 - 0.13} \\
 &= \frac{24}{29}
 \end{aligned}$$

Question 18 (4 marks) (J. Kim)

- ✓ [1] for applying trigonometric ratios of complementary angles or equivalent merit
- ✓ [2] for applying trigonometric identities or equivalent merit
- ✓ [3] for significant progress towards answer or equivalent merit
- ✓ [4] for correct solution

$$\begin{aligned}
 LHS &= \frac{\cos \theta}{\sin \theta} \times \sec^2 \theta + \frac{\sin \theta}{\cos \theta} \times \sec^2 \left(\frac{\pi}{2} - \theta \right) \\
 &= \frac{\cos \theta}{\sin \theta} \times \frac{1}{\cos^2 \theta} + \frac{\cos \theta}{\sin \theta} \times \operatorname{cosec}^2 \theta \\
 &= \frac{1}{\sin \theta \cos \theta} + \frac{\sin \theta}{\cos \theta} \times \frac{1}{\sin^2 \theta} \\
 &= \frac{1}{\sin \theta \cos \theta} + \frac{1}{\sin \theta \cos \theta} \\
 &= \frac{2}{\sin \theta \cos \theta} \\
 &= 2 \operatorname{cosec} \theta \sec \theta \\
 &= RHS
 \end{aligned}$$

Question 19 (3 marks) (J. Kim)

(a) (1 mark)

✓ [1] for correct answer

Since $t = 2$, $N = 2000$

$$\begin{aligned}1000 \times 1.6^{2k} &= 2000 \\1.6^{2k} &= 2 \\2k &= \frac{\ln 2}{\ln 1.6} \\k &= \frac{\ln 2}{2 \ln 1.6}\end{aligned}$$

(b) (2 marks)

✓ [1] for correct derivative or equivalent merit

✓ [2] for correct answer

$$\begin{aligned}\frac{dN}{dt} &= 1000 \times k \times 1.6^{kt} \times \ln 1.6 \\&= 1000k \ln(1.6) \times 1.6^{kt}\end{aligned}$$

When $t = 4$:

$$\begin{aligned}\frac{dN}{dt} &= 1000 \times \frac{\ln 2}{2 \ln 1.6} \times \ln 1.6 \times 1.6^{\frac{\ln 2}{2 \ln 1.6} \times 4} \\&= 1386.29 \dots \\&= 1386 \text{ tickets/week}\end{aligned}$$

Question 20 (6 marks) (Lam)

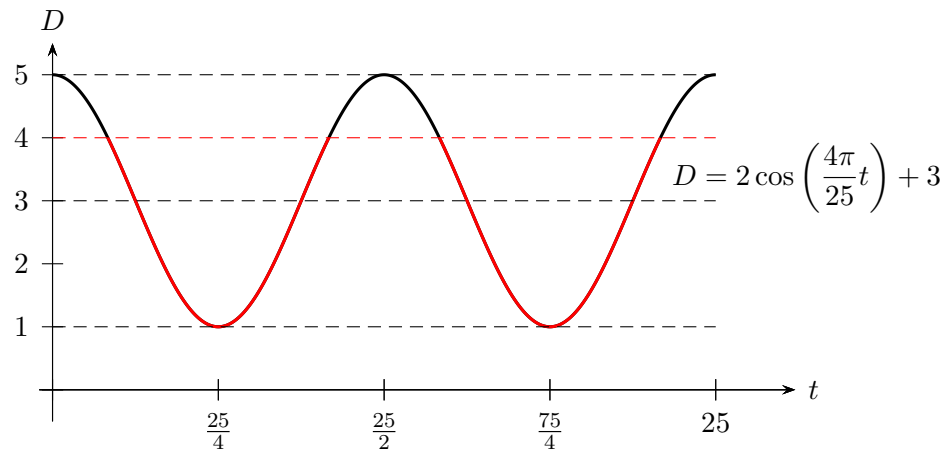
(a) (3 marks)

✓ [1] for one of the correct amplitude, or shape, or period, or equivalent merit.

✓ [2] for two of the correct amplitude, or shape, or period, or equivalent merit.

✓ [3] for correct solution.

$$\begin{aligned}T &= \frac{2\pi}{n} = \frac{2\pi}{\frac{4\pi}{25}} \\&= \frac{25}{2}\end{aligned}$$



(b) (3 marks)

- ✓ [1] for the correct value of $\cos\left(\frac{4\pi}{25}t\right)$, or equivalent merit. Shape includes the sketching of *two* periods of the graph.
- ✓ [2] for correct values of t , or equivalent merit.
- ✓ [3] for correct answer (absolute time values such as 0205 hrs not necessary for this question only)

$$2 \cos\left(\frac{4\pi}{25}t\right) + 3 = 4$$

$$\cos\left(\frac{4\pi}{25}t\right) = \frac{1}{2}$$

From the graph in (a), these points are the equivalent of the $\frac{\pi}{3}$ points on a non-transformed cosine graph, i.e.

$$\begin{aligned} \frac{4\pi}{25}t &= \frac{\pi}{3}, \frac{5\pi}{3}, 2\pi + \frac{\pi}{3}, 4\pi - \frac{\pi}{3} \\ &= \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3} \\ 4t &= \frac{25}{3}, \frac{125}{3}, \frac{175}{3}, \frac{275}{3} \\ t &= \frac{25}{12}, \frac{125}{12}, \frac{175}{12}, \frac{275}{12} \\ \therefore \frac{25}{12} < t < \frac{125}{12} \text{ or } \frac{175}{12} < t < \frac{275}{12} \end{aligned}$$

(Inequalities in t can be read off the graph).

Hence, the ship cannot cross between 0205 hrs to 1025 hrs, and 1435 hrs to 2055 hrs, exclusive at these timing points)

Question 21 (3 marks) (Lam)

- ✓ [1] for obtaining an integrand that is integrable by manipulation of the original integrand.
- ✓ [2] for correct primitive and substitution of limits, or equivalent merit.
- ✓ [3] for correct answer

$$\begin{aligned}
 \int_{-1}^1 \frac{1}{1+e^{-x}} \frac{\times e^x}{\times e^x} dx &= \int_{-1}^1 \frac{e^x}{1+e^x} dx \\
 &= \left[\ln |1+e^x| \right]_{-1}^1 \\
 &= \ln |1+e| - \ln |1+e^{-1}| \\
 &= \ln \left| \frac{1+e}{1+\frac{1}{e}} \right| \frac{\times e}{\times e} \\
 &= \ln \left| \frac{e(1+e)}{1+e} \right| \\
 &= \ln e = 1
 \end{aligned}$$

Alternatively, manipulate integrand as such:

$$\begin{aligned}
 \frac{1}{1+e^{-x}} &= \frac{1+e^{-x}-e^{-x}}{1+e^{-x}} \\
 &= \frac{1+e^{-x}}{1+e^{-x}} - \frac{e^{-x}}{1+e^{-x}} \\
 &= 1 + \frac{-e^{-x}}{1+e^{-x}}
 \end{aligned}$$

and then integrate to obtain $(x - \ln |1+e^{-x}|)$ as the primitive.

Question 22 (5 marks) (Lam)

(a) (3 marks)

- ✓ [1] for recognising a geometric series and subsequently, $\frac{T_3}{T_2} = \frac{T_2}{T_1}$ in terms of the cosine terms.
- ✓ [2] for finding $A = \frac{\pi}{6}$ (other potential solutions include $A = \frac{5\pi}{6}$ and also when $\cos A = -\frac{\sqrt{3}}{2}$, or equivalent merit.
- ✓ [3] for fully correct solution, with due consideration of the possible quadrant for $\angle A$ and also $T_1 > T_2 > T_3$.

$$\begin{aligned}\frac{\frac{1}{2}}{1 + \cos A} &= \frac{1 - \cos A}{\frac{1}{2}} \\ 1 - \cos^2 A &= \frac{1}{4} \\ \cos^2 A &= \frac{3}{4} \\ \cos A &= \pm \frac{\sqrt{3}}{2} \\ \therefore A &= \frac{\pi}{6}, \frac{5\pi}{6} \quad (\text{as } A < \pi)\end{aligned}$$

Since $T_1 > T_2 > T_3$, $\frac{T_2}{T_1} < 1$. If $A = \frac{\pi}{6}$,

$$\frac{\frac{1}{2}}{1 + \cos \frac{\pi}{6}} = \frac{1}{2 + \sqrt{3}} \approx 0.2679 < 1$$

If $A = \frac{5\pi}{6}$,

$$\begin{aligned}\frac{\frac{1}{2}}{1 + \cos \frac{5\pi}{6}} &= \frac{1}{2 - \sqrt{3}} \approx 3.73 > 1 \\ A &= \frac{\pi}{6} \text{ only}\end{aligned}$$

(b) (2 marks)

- ✓ [1] for utilising the value of A to find a value for the common ratio.
- ✓ [2] for fully showing the reason why the series has a limiting sum.

Since $A = \frac{\pi}{6}$,

$$\begin{aligned}T_1 &= 1 + \frac{\sqrt{3}}{2} & T_2 &= \frac{1}{2} \\ r &= \frac{\frac{1}{2}}{1 + \frac{\sqrt{3}}{2}} \\ &= \frac{\frac{1}{2}}{\frac{2 + \sqrt{3}}{2}} \\ &= \frac{1}{2 + \sqrt{3}} \\ &\approx 0.2679 \dots < 1\end{aligned}$$

Hence a limiting sum exists.

Question 23 (3 marks) (Lam)

- ✓ [1] for expansion of y^2 equation, or equivalent merit.
- ✓ [2] for the values of $T_3 - T_2$ and $T_2 - T_1$ in terms of y , m and n , or equivalent merit.
- ✓ [3] for correct solution. Note that students who automatically assumed $T_3 - T_2 = T_2 - T_1$ can only receive 2 out of 3 marks maximum, as they assumed the terms were equal prior to showing the terms form an arithmetic sequence.

$$\begin{aligned}
 y^2 &= (2m - y)(2n - y) \\
 y^2 &= 4mn - 2my - 2ny + y^2 \\
 2my + 2ny &= 4mn \\
 y &= \frac{2mn}{m + n} \\
 \frac{1}{y} &= \frac{m + n}{2mn}
 \end{aligned}$$

Inspecting $T_2 - T_1$:

$$\begin{aligned}
 T_2 - T_1 &= \frac{1}{y} - \frac{1}{m} \\
 &= \frac{m + n}{2mn} - \frac{1}{m} \times \frac{2n}{2n} \\
 &= \frac{m + n - 2n}{2n} \\
 &= \frac{m - n}{2n}
 \end{aligned}$$

Inspecting $T_3 - T_2$:

$$\begin{aligned}
 T_3 - T_2 &= \frac{1}{n} - \frac{1}{y} \\
 &= \frac{1}{n} \times \frac{2m}{2m} - \frac{m + n}{2mn} \\
 &= \frac{2m - m - n}{2mn} \\
 &= \frac{m - n}{2mn} = T_2 - T_1
 \end{aligned}$$

As $T_3 - T_2 = T_2 - T_1$, the three terms form an arithmetic sequence.

Question 24 (6 marks) (Lam)

(a) (4 marks)

- ✓ [1] for the correct value of AO .
- ✓ [2] for applying cosine rule to $\triangle OAB$, or equivalent merit.
- ✓ [3] for forming a quadratic equation in terms of OB .
- ✓ [4] for correct answer.

In $\triangle AOT$:

$$\begin{aligned}
 \cos 60^\circ &= \frac{AO}{3\sqrt{3}} \\
 AO &= \frac{3\sqrt{3}}{2}
 \end{aligned}$$

From the bearing provided,

$$\angle AOB = 270^\circ - 240^\circ = 30^\circ$$

Applying the cosine rule in $\triangle AOB$:

$$\begin{aligned} 10^2 &= \left(\frac{3\sqrt{3}}{2}\right)^2 + (OB)^2 - 2\left(\frac{3\sqrt{3}}{2}\right)(OB)\cos 30^\circ \\ \frac{27}{4} + (OB)^2 - 3\sqrt{3} \times \frac{\sqrt{3}}{2}(OB) &= 100 \\ (OB)^2 - \frac{9}{2}(OB) - \frac{373}{4} &= 0 \\ 4(OB)^2 - 18(OB) - 373 &= 0 \\ OB &= \frac{18 \pm \sqrt{18^2 - 4 \times 4 \times (-373)}}{2 \times 4} \\ &= \frac{18 \pm \sqrt{6292}}{8} \\ &= \frac{18 \pm 22\sqrt{13}}{8} \\ &= \frac{9 \pm 11\sqrt{13}}{4} \end{aligned}$$

Since $OB > 0$,

$$OB = \frac{9 + 11\sqrt{13}}{4} \text{ metres}$$

(b) (2 marks)

- ✓ [1] for the correct value of OT .
- ✓ [2] for correct solution.

$$\begin{aligned} \sin 60^\circ &= \frac{OT}{3\sqrt{3}} \\ OT &= \frac{9}{2} \\ \ell^2 &= OB^2 + OT^2 \\ &= \left(\frac{9 + 11\sqrt{13}}{4}\right)^2 + \left(\frac{9}{2}\right)^2 \\ \ell &= \pm \sqrt{\left(\frac{9 + 11\sqrt{13}}{4}\right)^2 + \left(\frac{9}{2}\right)^2} \\ &= 12.97 \dots \quad (\ell > 0) \\ &= 13 \text{ m} \end{aligned}$$

Question 25 (3 marks) (Chaussivert)

- ✓ [1] for finding m in terms of x or equivalent merit
- ✓ [2] for finding x in terms of k or equivalent merit
- ✓ [3] for correct answer

$$\begin{aligned}
 y &= \ln(kx) \\
 \frac{dy}{dx} &= \frac{k}{kx} \\
 &= \frac{1}{x} \\
 m &= \frac{1}{x}
 \end{aligned}$$

Since $y = mx$ and $y = \ln(kx)$ intersect:

$$\begin{aligned}
 mx &= \ln(kx) \\
 \text{Sub } m &= \frac{1}{x} \\
 \ln(kx) &= \frac{1}{x} \times x \\
 \ln(kx) &= 1 \\
 kx &= e \\
 x &= \frac{e}{k} \\
 \therefore m &= \frac{1}{\frac{e}{k}} \\
 &= \frac{k}{e}
 \end{aligned}$$

Question 26 (6 marks) (Chaussivert)

(a) (3 marks)

- ✓ [1] for factorisation of exponential function or equivalent merit
- ✓ [2] for finding x or equivalent merit
- ✓ [3] for correct answer

$$\begin{aligned}
 e^x - 1 &= 2e^{-x} \\
 e^{2x} - e^x - 2 &= 0 \\
 (e^x - 2)(e^x + 1) &= 0 \\
 e^x &= -1 \text{ or } 2 \\
 e^x &= 2 \quad (e^x > 0) \\
 x &= \ln 2 \\
 y &= e^{\ln 2} - 1 \\
 &= 1 \\
 \therefore P(\ln 2, 1)
 \end{aligned}$$

(b) (3 marks)

- ✓ [1] for correct expression of the area or equivalent merit
- ✓ [2] for correct integration or equivalent merit
- ✓ [3] for correct answer

$$\begin{aligned}
 A &= \int_0^{\ln 2} 2e^{-x} - e^x + 1 \, dx \\
 &= \left[-2e^{-x} - e^x + x \right]_0^{\ln 2} \\
 &= -2e^{-\ln 2} - e^{\ln 2} + \ln 2 - (-2e^0 - e^0 + 0) \\
 &= -2\left(\frac{1}{2}\right) - 2 + \ln 2 - (-2 - 1) \\
 &= \ln 2 - 3 + 3 \\
 &= \ln 2
 \end{aligned}$$

Question 27 (8 marks) (Chaussivert)

(a) (2 marks)

- ✓ [1] for use of trapezoidal rule or equivalent merit
- ✓ [2] for correct answer

x	1	3	5	7
$f(x)$	0	$\ln 3$	$\ln 5$	$\ln 7$

$$\begin{aligned}
 h &= \frac{7-1}{3} \\
 &= 2 \\
 A &= \frac{2}{2} [0 + \ln 7 + 2(\ln 3 + \ln 5)] \\
 &= 2 \ln 3 + 2 \ln 5 + \ln 7
 \end{aligned}$$

(b) (2 marks)

- ✓ [1] for progress towards answer
- ✓ [2] for correct answer

$$\begin{aligned}
 \frac{d}{dx} (x \ln x) &= \ln x + \frac{1}{x} \times x \\
 &= \ln x + 1
 \end{aligned}$$

(c) (3 marks)

- ✓ [1] for integrating both sides of equation in part (b) or equivalent merit
- ✓ [2] for correct integration or equivalent merit
- ✓ [3] for correct answer

From part (b):

$$\begin{aligned}
 \frac{d}{dx}(x \ln x) &= \ln x + 1 \\
 \int_1^7 \ln x + 1 \, dx &= \int_1^7 \frac{d}{dx}(x \ln x) \\
 \int_1^7 \ln x \, dx + \int_1^7 dx &= [x \ln x]_1^7 \\
 \int_1^7 \ln x \, dx &= [x \ln x]_1^7 - [x]_1^7 \\
 &= 7 \ln 7 - 0 - (7 - 1) \\
 &= 7 \ln 7 - 6
 \end{aligned}$$

(d) (1 mark)

- ✓ [1] for correct answer

Equating part (a) and (c):

$$\begin{aligned}
 7 \ln 7 - 6 &= 2 \ln 3 + 2 \ln 5 + \ln 7 \\
 6 \ln 7 &= 2 \ln 3 + 2 \ln 5 + 6 \\
 \ln 7 &\approx \frac{1}{3} (\ln 3 + \ln 5) + 1
 \end{aligned}$$

Question 28 (7 marks) (Lee)

(a) (2 marks)

- ✓ [1] for integration of acceleration or equivalent merit
- ✓ [2] for correct answer

$$\begin{aligned}
 \dot{x} &= \int 4 \sin 2t \, dt \\
 &= -2 \cos 2t + C
 \end{aligned}$$

When $t = 0$, $\dot{x} = -1$

$$\begin{aligned}
 -1 &= -2 \cos 0 + C \\
 C &= 1 \\
 \therefore \dot{x} &= 1 - 2 \cos 2t
 \end{aligned}$$

(b) (2 marks)

- ✓ [1] for finding $\cos 2t$ or equivalent merit
- ✓ [2] for correct answer

When $\dot{x} = 0$, $t = ?$

$$1 - 2 \cos 2t = 0$$

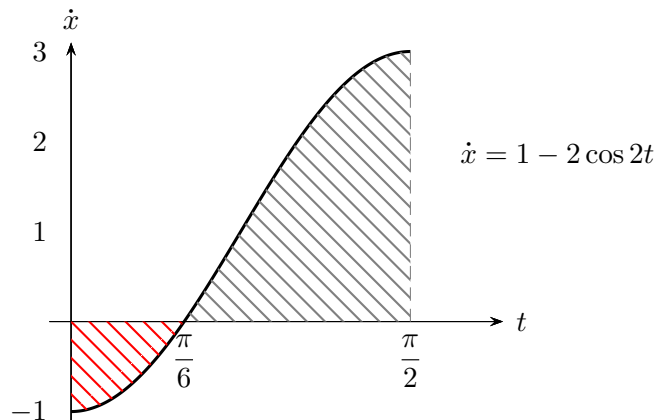
$$\cos 2t = \frac{1}{2}$$

$$2t = \frac{\pi}{3}$$

$$t = \frac{\pi}{6} \text{ seconds}$$

(c) (3 marks)

- ✓ [1] for expression of distance or equivalent merit
- ✓ [2] for correct integration or equivalent merit
- ✓ [3] for correct answer



$$\begin{aligned}
 d &= \left| \int_0^{\frac{\pi}{6}} 1 - 2 \cos 2t \, dt \right| + \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} 1 - 2 \cos 2t \, dt \\
 &= \left| t - \sin 2t \right|_0^{\frac{\pi}{6}} + \left[t - \sin 2t \right]_{\frac{\pi}{6}}^{\frac{\pi}{2}} \\
 &= \left| \frac{\pi}{6} - \sin \frac{\pi}{3} - 0 \right| + \frac{\pi}{2} - \sin \pi - \left(\frac{\pi}{6} - \sin \frac{\pi}{3} \right) \\
 &= \left| \frac{\pi}{6} - \frac{\sqrt{3}}{2} \right| + \frac{\pi}{2} - \left(\frac{\pi}{6} - \frac{\sqrt{3}}{2} \right) \\
 &= \frac{\sqrt{3}}{2} - \frac{\pi}{6} + \frac{\pi}{2} - \frac{\pi}{6} + \frac{\sqrt{3}}{2} \\
 &= \sqrt{3} + \frac{\pi}{6} \text{ metres}
 \end{aligned}$$

Question 29 (7 marks) (Lee)

(a) (3 marks)

- ✓ [1] for finding QR or equivalent merit
- ✓ [2] for finding y or equivalent merit
- ✓ [3] for correct answer

In $\triangle QOR$:

$$\tan \theta = \frac{QR}{x}$$

$$QR = 2x$$

In $\triangle POS$:

$$(x+y)^2 + PS^2 = OP^2$$

$$(x+y)^2 + (2x)^2 = 5^2$$

$$(x+y)^2 = 25 - 4x^2$$

Since $x+y > 0$

$$x+y = \sqrt{25-4x^2}$$

$$y = -x + \sqrt{25-4x^2}$$

$$P = 2(2x) + 2(-x + \sqrt{25-4x^2})$$

$$= 2x + 2\sqrt{25-4x^2}$$

(b) (4 marks)

- ✓ [1] for finding $\frac{dP}{dx}$ or equivalent merit
- ✓ [2] for finding x or equivalent merit
- ✓ [3] for showing maximum at x or equivalent merit
- ✓ [4] for correct answer

$$P = 2x + 2(25 - 4x^2)^{\frac{1}{2}}$$

$$\frac{dP}{dx} = 2 + 2 \times \frac{1}{2} (25 - 4x^2)^{-\frac{1}{2}} \times -8x$$

$$= 2 - \frac{8x}{\sqrt{25 - 4x^2}}$$

Let $\frac{dP}{dx} = 0$

$$2 - \frac{8x}{\sqrt{25 - 4x^2}} = 0$$

$$\frac{8x}{\sqrt{25 - 4x^2}} = 2$$

$$4x = \sqrt{25 - 4x^2}$$

$$16x^2 = 25 - 4x^2$$

$$20x^2 = 25$$

$$x^2 = \frac{5}{4}$$

$$x = \frac{\sqrt{5}}{2} \quad (x > 0)$$

x	1	$\frac{\sqrt{5}}{2}$	2
$\frac{dP}{dx}$	0.25	0	$-\frac{10}{3}$
P	$\nearrow$	$-$	$\searrow$

$\therefore$ maximum at $x = \frac{\sqrt{5}}{2}$

$$P = 2 \times \frac{\sqrt{5}}{2} + 2\sqrt{25 - 4\left(\frac{\sqrt{5}}{2}\right)^2}$$

$$= \sqrt{5} + 2\sqrt{20}$$

$$= 5\sqrt{5} \text{ metres}$$